

A quasi-polynomial q -Queens result and related Kronecker products of matrices

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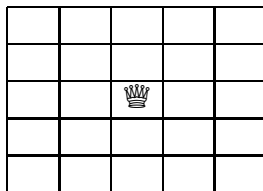
Joint work with Seth Chaiken, [University at Albany](#)
and Tom Zaslavsky, [Binghamton University](#)

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The n -Queens Problem

Motivating question:

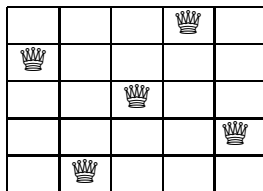
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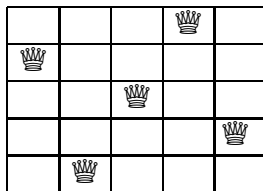


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n	1	2	3	4	5	6	7	8	9	10
#	1	0	0	2	10	4	40	92	352	724

The q -Queens Problem and generalizations

Let's generalize.

- ▶ Fix the number of queens. (q)
- ▶ Let the size of the board vary. ($n \times n$)

Question: Determine the number of ways in which you can place q nonattacking queens on an $n \times n$ chessboard as a function of n .

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Question: Why stop there?

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A problem will have three elements:

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- ▶ A board. (A convex polygon and its dilations.)
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 Nightrider: $\mathbf{M} = \{(2, 1), (1, 2), (2, -1), (1, -2)\}$

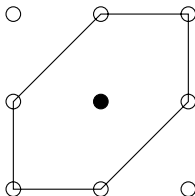
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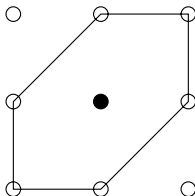


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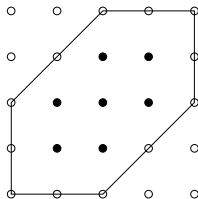


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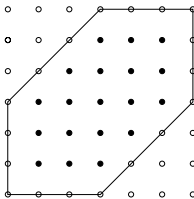


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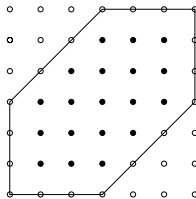


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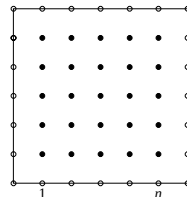
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The $n \times n$ case corresponds to $t = (n + 1)$.



The q -Queens Problem and generalizations

Theorem: (Chaiken, Zaslavsky, 2005)

Given P , $\mathcal{B}$, and q , the number of placements of q nonattacking P pieces inside $t\mathcal{B}$ is a quasipolynomial function of t .

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A *quasipolynomial* is a function $f(t)$ on $t \in \mathbb{Z}_+$ such that

$$f(t) = c_d t^d + c_{d-1} t^{d-1} + \cdots + c_0,$$

where each c_i is periodic.

Example:
$$f(t) = \begin{cases} t^2 + 3t + 2 & \text{for even } t \\ t^2 - 2t + 1 & \text{for odd } t \end{cases}$$

q -Queens proof sketch

Very briefly:

The rules of nonattack correspond to forbidden hyperplanes in $\mathbb{R}^{2q}$.

Inside-out polytope theory gives a quasipolynomial function of t .

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- ▶ Count lattice points in $t\mathcal{B}^q$ avoiding $\mathcal{H}$.

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- ▶ There are $\binom{q}{2}N$ of these forbidden hyperplanes in $\mathbb{R}^{2q}$
- ▶ Count lattice points in $t\mathcal{B}^q$ avoiding $\mathcal{H}$.
- ▶ This is a direct application of inside-out polytope theory
Counted by a quasipolynomial with certain properties.

Inside-out polytopes

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- ▶ Builds upon ideas of Ehrhart theory.
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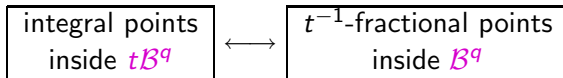
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Therefore: The number nonattacking configurations of q pieces $\mathcal{P}$ inside $t\mathcal{B}$ is a quasipolynomial function of t with

- ▶ degree: $\dim(\mathcal{B}^q) = 2q$.
- ▶ leading coefficient: $|\mathcal{B}|^q/q!$. ← Now unordered!

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- ▶ Determine the period of the coefficients.
- ▶ Compute initial data to determine the formula.

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Notation: Write $u_P(q; n)$ for the number of (unlabeled) nonattacking configurations of q pieces P on an $n \times n$ board.

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For a fixed q , we expect a formula of the form:

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Classic result for rooks R :

$$\text{♖} \quad u_R(q; n) = q! \binom{n}{q}^2.$$

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- ▶ $u_B(2; n) = \frac{1}{2}n^4 + c_3n^3 + c_2n^2 + c_1n + c_0$.
- ▶ Initial data for $u_B(2; n)$:

n	1	2	3	4	5	6	7	8
$u_B(2; n)$	0	4	26	92	240	520	994	1736

$$u_B(2; n) = \frac{1}{2}n^4 - \frac{2}{3}n^3 + \frac{1}{2}n^2 - \frac{1}{3}n.$$

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 (CHZ 20??) For bishops B , the period divides 2.

$$u_B(1; n) = n^2.$$

$$u_B(2; n) = \frac{n^4}{2} - \frac{2n^3}{3} + \frac{n^2}{2} - \frac{n}{3}$$

$$u_B(3; n) = \left\{ \frac{n^6}{6} - \frac{2n^5}{3} + \frac{5n^4}{4} - \frac{5n^3}{3} + \frac{4n^2}{3} - \frac{2n}{3} + \frac{1}{8} \right\} - (-1)^n \frac{1}{8}.$$

$$u_B(4; n) = \left\{ \frac{n^8}{24} - \frac{n^7}{3} + \frac{11n^6}{9} - \frac{29n^5}{10} + \frac{355n^4}{72} - \frac{35n^3}{6} + \frac{337n^2}{72} - \frac{73n}{30} + \frac{1}{2} \right\} - (-1)^n \left\{ \frac{n^2}{8} - \frac{n}{2} + \frac{1}{2} \right\}.$$

$$u_B(5; n) = \left\{ \frac{n^{10}}{120} - \frac{n^9}{9} + \frac{49n^8}{72} - \frac{118n^7}{45} + \frac{523n^6}{72} - \frac{2731n^5}{180} + \frac{3413n^4}{144} - \frac{4853n^3}{180} + \frac{2599n^2}{120} - \frac{1321n}{120} + \frac{9}{4} \right\} - (-1)^n \left\{ \frac{n^4}{16} - \frac{7n^3}{12} + \frac{17n^2}{8} - \frac{85n}{24} + \frac{9}{4} \right\}.$$

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Finding the quasipolynomial period

$$\begin{pmatrix} M & -M & 0 & 0 & \cdots & 0 & 0 \\ M & 0 & -M & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ M & 0 & 0 & 0 & \cdots & 0 & -M \\ 0 & M & -M & 0 & \cdots & 0 & 0 \\ 0 & M & 0 & -M & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & M & 0 & 0 & \cdots & 0 & -M \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & M & -M \\ \hline B & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & B & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & B \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_q \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ \hline \beta \\ \beta \\ \vdots \\ \beta \end{pmatrix}$$

where M is the
moves matrix

$$M := \begin{pmatrix} m_1^+ \\ m_1^- \\ m_2^+ \\ \vdots \\ m_{|M|}^+ \end{pmatrix}$$

and

$$B := \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \\ \vdots & \vdots \\ a_K & b_K \end{pmatrix}.$$

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$$\begin{pmatrix} M & -M & 0 & 0 & \cdots & 0 & 0 \\ M & 0 & -M & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ M & 0 & 0 & 0 & \cdots & 0 & -M \\ 0 & M & -M & 0 & \cdots & 0 & 0 \\ 0 & M & 0 & -M & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & M & 0 & 0 & \cdots & 0 & -M \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & M & -M \\ \hline B & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & B & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & B \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_q \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ \hline \beta \\ \beta \\ \vdots \\ \beta \end{pmatrix}$$

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► Cramer's Rule $\rightsquigarrow$ vertex denominator divides a subdet. of A .

Finding the quasipolynomial period

$$\begin{pmatrix} M & -M & 0 & 0 & \cdots & 0 & 0 \\ M & 0 & -M & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ M & 0 & 0 & 0 & \cdots & 0 & -M \\ 0 & M & -M & 0 & \cdots & 0 & 0 \\ 0 & M & 0 & -M & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & M & 0 & 0 & \cdots & 0 & -M \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & M & -M \\ \hline B & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & B & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & B \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_q \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ \hline \beta \\ \beta \\ \vdots \\ \beta \end{pmatrix}$$

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$$M := \begin{pmatrix} m_1^\perp \\ m_2^\perp \\ \vdots \\ m_{|M|}^\perp \end{pmatrix}$$

and

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- ▶ Cramer's Rule $\rightsquigarrow$ vertex denominator divides a subdet. of A .
- ▶ Period of quasipoly. divides **lcm of all such subdet's**, $\text{lcmd}(A)$.

Finding the quasipolynomial period

$$\begin{pmatrix}
 M & -M & 0 & 0 & \cdots & 0 & 0 \\
 M & 0 & -M & 0 & \cdots & 0 & 0 \\
 \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\
 M & 0 & 0 & 0 & \cdots & 0 & -M \\
 0 & M & -M & 0 & \cdots & 0 & 0 \\
 0 & M & 0 & -M & \cdots & 0 & 0 \\
 \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\
 0 & M & 0 & 0 & \cdots & 0 & -M \\
 \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\
 0 & 0 & 0 & 0 & \cdots & M & -M \\
 \hline
 B & 0 & 0 & 0 & \cdots & 0 & 0 \\
 0 & B & 0 & 0 & \cdots & 0 & 0 \\
 \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\
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where M is the moves matrix

$$M := \begin{pmatrix} m_1^\perp \\ m_2^\perp \\ \vdots \\ m_{|M|}^\perp \end{pmatrix}$$

and

$$B := \begin{pmatrix} -1 & 0 \\ 0 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \begin{array}{l} -x \leq 0 \\ -y \leq 0 \\ x \leq 1 \\ y \leq 1 \end{array}$$

- ▶ Cramer's Rule $\rightsquigarrow$ vertex denominator divides a subdet. of A .
- ▶ Period of quasipoly. divides **lcm of all such subdet's**, $\text{lcmd}(A)$.
- ▶ A square board simplifies.

Finding the quasipolynomial period

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 M & 0 & 0 & 0 & \cdots & 0 & -M \\
 0 & M & -M & 0 & \cdots & 0 & 0 \\
 0 & M & 0 & -M & \cdots & 0 & 0 \\
 \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\
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- ▶ Cramer's Rule $\rightsquigarrow$ vertex denominator divides a subdet. of A .
- ▶ Period of quasipoly. divides **lcm of all such subdet's**, $\text{lcmd}(A)$.
- ▶ A square board simplifies. The structure of A' is predictable.



The structure of A'

$$A' = \begin{pmatrix} M & -M & 0 & 0 & \cdots & 0 & 0 \\ M & 0 & -M & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ M & 0 & 0 & 0 & \cdots & 0 & -M \\ 0 & M & -M & 0 & \cdots & 0 & 0 \\ 0 & M & 0 & -M & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & M & 0 & 0 & \cdots & 0 & -M \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & M & -M \end{pmatrix}$$

The structure of A'

$$(A')^T = \begin{pmatrix} M^T & M^T & \cdots & M^T & 0 & 0 & \cdots & 0 & \cdots & 0 \\ -M^T & 0 & \cdots & 0 & M^T & M^T & \cdots & M^T & \cdots & 0 \\ 0 & -M^T & \cdots & 0 & -M^T & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & -M^T & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \cdots & M^T \\ 0 & 0 & \cdots & -M^T & 0 & 0 & \cdots & -M^T & \cdots & -M^T \end{pmatrix}$$

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This reminds us of the incidence matrix for the complete graph K_q :

$$D(K_q) = \begin{pmatrix} 1 & 1 & \cdots & 1 & 0 & 0 & \cdots & 0 & \cdots & 0 \\ -1 & 0 & \cdots & 0 & 1 & 1 & \cdots & 1 & \cdots & 0 \\ 0 & -1 & \cdots & 0 & -1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & -1 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \cdots & 1 \\ 0 & 0 & \cdots & -1 & 0 & 0 & \cdots & -1 & \cdots & -1 \end{pmatrix}$$

The structure of F^T

For matrices $A = (a_{ij})_{m \times k}$ and $B = (b_{ij})_{n \times l}$, the *Kronecker product* $A \otimes B$ is defined to be the $mn \times kl$ block matrix

$$\begin{bmatrix} a_{11}B & \cdots & a_{1k}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mk}B \end{bmatrix}.$$

The structure of F^T

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We have $(A')^T = D(K_q) \otimes M^T$

- ▶ M is the $m \times 2$ moves matrix
- ▶ $D(K_q)$ is the incidence matrix for the complete graph K_q .

About Kronecker Products

About Kronecker products:

- ▶ $A \otimes B$ and $B \otimes A$ only differ by row and column switchings.
- ▶ For $A_{m \times m}$ and $B_{n \times n}$, $\det(A \otimes B) = \det(A)^n \det(B)^m$.

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- ▶ We aim to simplify $\text{lcmd}(M^T \otimes D(K_q))$.
- ▶ Funny story.

lcmd result

Theorem (Hanusa, Zaslavsky, 2008) Given $M_{m \times 2}$ and $q \geq 1$,

$$\text{lcmd} (M \otimes D(K_q)) = \text{lcm} \left((\text{lcmd } M)^{q-1}, \text{LCM}_{\mathcal{K}} \left(\prod_{(I,J) \in \mathcal{K}} \det M^{I,J} \right) \right),$$

The LCM is over disjoint multisubsets I and J of $[m]$ of size $\lfloor q/2 \rfloor \dots$

► $\text{lcmd} (M \otimes D(K_q))$ is simply an lcm over entries of M .

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Example: For $M_{4 \times 2}$, we have $m = 4$.

Find all pairs (I, J) of disjoint $\lfloor q/2 \rfloor$ -member multisubsets of $[4]$:

$$\begin{aligned} &(\{1^a\}, \{2^b, 3^c, 4^d\}), \quad (\{2^b\}, \{1^a, 3^c, 4^d\}), \\ &(\{3^c\}, \{1^a, 2^b, 4^d\}), \quad (\{4^d\}, \{1^a, 2^b, 3^c\}), \\ &(\{1^a, 2^b\}, \{3^c, 4^d\}), \quad (\{1^a, 3^c\}, \{2^b, 4^d\}), \quad (\{1^a, 4^d\}, \{2^b, 3^c\}), \end{aligned}$$

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Example: For $M_{4 \times 2}$, $m = 4$. Consider $(I, J) = (\{1^a, 2^b\}, \{3^c, 4^d\})$.

$$\text{Then, } M^{I,J} = \begin{pmatrix} m_{11}^a m_{21}^b & m_{12}^a m_{22}^b \\ m_{31}^c m_{41}^d & m_{32}^c m_{42}^d \end{pmatrix},$$

for all a, b, c , and d such that $a + b = c + d = \lfloor q/2 \rfloor$.

Bishop example

Example: When $P = \text{♞}$ (bishop),

$$M = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \text{ has } \text{lcmd}(M) = 2.$$

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$$\blacktriangleright (I, J) = (\{1^p\}, \{2^p\}).$$

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- ▶ And our quasipolynomial period must divide 2^{q-1}

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- ▶ The LCM term generates powers of 2 no larger than $2^{q/2}$.
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Back to that funny story...

Sketch of Kronecker theorem proof

$$\text{lcmd} (M \otimes D(K_q)) = \text{lcm} \left((\text{lcmd } M)^{q-1}, \text{LCM}_{\mathcal{K}} \left(\prod_{(I,J) \in \mathcal{K}} \det M^{I,J} \right) \right),$$

Goal: Show that every $N_{I \times I}$ subdet. of $M \otimes D(K_q)$ divides RHS.

- ▶ Consider only N such that $\det(N) \neq 0$.
- ▶ N is a choice of I rows and I columns from $M \otimes D(K_q)$
- ▶ Same as a choice of I vertices and I edges from K_q , with up to m copies of each vertex and up to two copies of an edge.

$$M \otimes D(K_q) = \begin{pmatrix} m_{11} D(K_q) & m_{12} D(K_q) \\ m_{21} D(K_q) & m_{22} D(K_q) \\ \vdots & \vdots \\ m_{m1} D(K_q) & m_{m2} D(K_q) \end{pmatrix}$$

Sketch of Kronecker theorem proof

- ▶ When two rows correspond to the same vertex v , the rows contain the same entries except for different multipliers m_{ik} .

$$\begin{pmatrix} m_{11}D(K_q) & m_{12}D(K_q) \\ m_{21}D(K_q) & m_{22}D(K_q) \\ \vdots & \vdots \\ m_{m1}D(K_q) & m_{m2}D(K_q) \end{pmatrix} \begin{pmatrix} \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots & \vdots \\ m_{j1} & -m_{j1} & 0 & \cdots & 0 & m_{j2} & 0 & \cdots & -m_{j2} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots & \vdots \\ m_{j1} & -m_{j1} & 0 & \cdots & 0 & m_{j2} & 0 & \cdots & -m_{j2} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots & \vdots \end{pmatrix}$$

- ▶ A vertex chosen three or more times would imply lin. dep.
- ▶ Simplify $\det N$ when a vertex is chosen twice.
(This generates a factor of $\det M^{i,j}$).

$$\begin{pmatrix} \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots & \vdots \\ m_{j1} & -m_{j1} & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & m_{j2} & 0 & \cdots & -m_{j2} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots & \vdots \end{pmatrix}$$

Sketch of Kronecker theorem proof

- ▶ Afterwards, every column has at most two entries.
- ▶ If a row (or column) has exactly one non-zero entry, expand.
- ▶ Then every row has exactly two non-zero entries as well.
- ▶ This matrix breaks down as a product of incidence matrices of weighted cycles, each of which basically contributes $\det M^{I,J}$.

$$\begin{pmatrix} y_1 & 0 & 0 & 0 & 0 & -z_6 \\ -z_1 & y_2 & 0 & 0 & 0 & 0 \\ 0 & -z_2 & y_3 & 0 & 0 & 0 \\ 0 & 0 & -z_3 & y_4 & 0 & 0 \\ 0 & 0 & 0 & -z_4 & y_5 & 0 \\ 0 & 0 & 0 & 0 & -z_5 & y_6 \end{pmatrix}.$$

Not Queens

When $M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 1 & -1 \end{pmatrix}$, again $\text{lcmd}(M) = 2$.

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Calculate $\det M^{I,J}$ for each pair (I, J) :

- For example, when $I = \{3^c\}$ and $J = \{1^a, 2^b, 4^d\}$, and

$$M^{I,J} = \begin{pmatrix} 1^c & 1^c \\ 1^a 0^b 1^d & 0^a 1^b (-1)^d \end{pmatrix}.$$

where $c = a + b + d = \lfloor q/2 \rfloor$.

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- ▶ The only non-trivial case is when $a = b = 0$. Therefore $c = d = \lfloor q/2 \rfloor$ and $\det M^{I,J} = 0$ or -2 .
- ▶ This implies that the LCM in the theorem divides 2^{q-1} .

We conclude that $\text{lcmd}(M \otimes D(K_q)) = 2^{q-1}$.

Not Nightriders

Consider $M = \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 1 & -2 \\ 2 & -1 \end{pmatrix}$.

- The submatrices

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 1 & -2 \end{pmatrix}, \text{ and } \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$$

have determinants -3 , -4 , and -5 ; hence $\text{lcmd}(M) = 60$.

- We have the same multisubsets of $[4]$ as before.
- $\det M^{I,J}$ is the same form in all cases: $\pm 2^u(2^{2\lfloor q/2 \rfloor - 2u} \pm 1)$, where u is a number between 0 and $\lfloor q/2 \rfloor$.

Not Nightriders

We conclude that

$$\text{lcmd} \left(M \otimes D(K_q) \right) = \text{lcm} \left(60^{q-1}, \text{LCM}_{\substack{1 \leq p \leq q/2 \\ 0 \leq u \leq p-1}} (2^{2p-2u} \pm 1)^{\lfloor q/2p \rfloor} \right).$$

When $q = 8$, $\text{lcmd} \left(M \otimes D(K_8) \right) =$

$$\begin{aligned} & \text{lcm} \left(60^7, (4 \pm 1)^{\lfloor 8/2 \rfloor}, (16 \pm 1)^{\lfloor 8/4 \rfloor}, (64 \pm 1)^{\lfloor 8/6 \rfloor}, (256 \pm 1)^{\lfloor 8/8 \rfloor} \right) \\ &= 60^7 \cdot 7 \cdot 13 \cdot 17^2 \cdot 257. \end{aligned}$$

Not Nightriders

The first few values of q give the following numbers:

q	$\text{lcmd} (M \otimes D(K_q))$	(factored)
2	60	60^1
3	3600	60^2
4	3672000	$60^3 \cdot 17$
5	220320000	$60^4 \cdot 17$
6	1202947200000	$60^5 \cdot 7 \cdot 13 \cdot 17$
7	72176832000000	$60^6 \cdot 7 \cdot 13 \cdot 17$
8	18920434740480000000	$60^7 \cdot 7 \cdot 13 \cdot 17^2 \cdot 257$
9	1135226084428800000000	$60^8 \cdot 7 \cdot 13 \cdot 17^2 \cdot 257$
10	952295753183943168000000000	$60^9 \cdot 7 \cdot 11 \cdot 13 \cdot 17^2 \cdot 31 \cdot 41 \cdot 257$

View of our wandering from above

- ▶ Generalize n -Queens to q -Queens and beyond.
- ▶ Apply inside-out polytope theory to prove formula existence.
- ▶ Need to know the period; aim to find $\text{lcmd}(A)$.

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- ▶ Generalize n -Queens to q -Queens and beyond.
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- ▶ On a rectangular board, $\text{lcmd}(A) = \text{lcmd}(M^T \otimes D(K_q))$.
- ▶ Prove a theorem that applies to find $\text{lcmd}(M \otimes D(K_q))$.
- ▶ The theorem applies for $M_{2 \times 2}$.

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$$\text{lcmd} (M \otimes D(K_q)) = \text{lcm} \left((\text{lcmd } M)^{q-1}, \text{LCM}_{\mathcal{K}} \left(\prod_{(I,J) \in \mathcal{K}} \det M^{I,J} \right) \right),$$

Open problems

- ▶ A better way to find the period? (LCMD is “bad”)
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$p \cdot q$ When do two multivariate binomials have a common divisor?
 $(wx^2y - z^2u^2)$ and $(wy^3 - xz^2u)$

Thank you

Slides available: people.qc.cuny.edu/chanusa > Talks



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